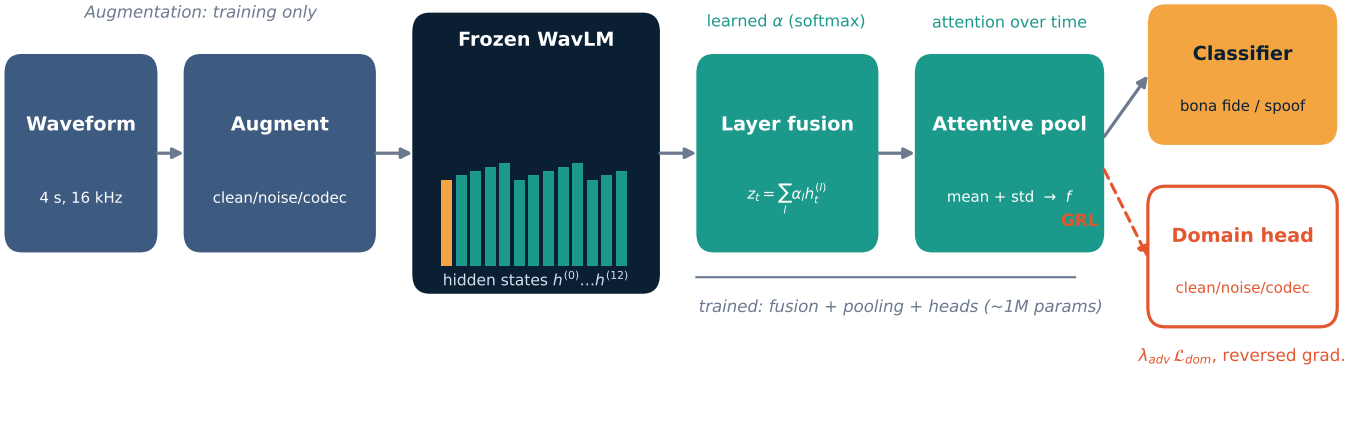
$\mathcal{L}_{cls}$: weighted CE
Augmentation: training only
Waveform
4 s, 16 kHz
Augment
clean/noise/codec
Frozen WavLM
hidden states $h^{(0)}…h^{(12)}$
learned α (softmax)
Layer fusion
$z_t = \sum_l \alpha_l h_t^{(l)}$
attention over time
Attentive pool
mean + std → f
GRL
Classifier
bona fide / spoof
Domain head
clean/noise/codec
trained: fusion + pooling + heads (~1M params)
$\lambda_{adv}\,\mathcal{L}_{dom}$, reversed grad.